Design of An Anatomically Correct Dynamic Phantom for the TAVR Procedure

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Introduction

What is TAVR & Why it Matters?

TAVR is a minimally invasive procedure that replaces a narrowed aortic valve to restore blood flow [1].

- Catheter delivery (leg artery)
- Valve expansion & function replacement
- No open-heart → faster recovery (high-risk patients) [2]

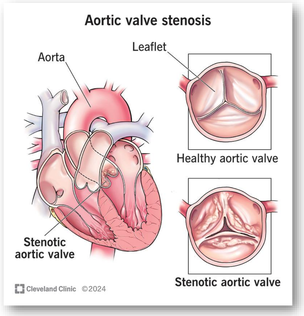


Figure 1 Illustration of Aortic Valve Stenosis

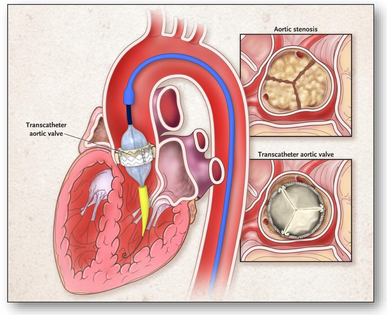


Figure 2 Illustration of TAVR Procedure

Digital Twin in TAVR

A digital twin is a CT-based, patient-specific computational model that predicts aortic and valve behavior during TAVR.

- CT → Digital model
- Predicts valve/aorta behavior
- Validated by physical phantom

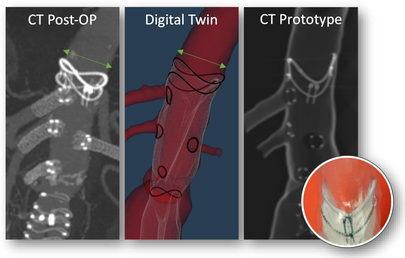


Figure 3 CT Scan to Digital Simulation

Figure 4 Fabricated Phantom

Design Objectives

Design Goal:

Develop an integrated system to validate CT-based digital twin predictions for TAVR under realistic flow conditions.

Components and Design Requirements:

Aortic Phantom

- Reflecting realistic aortic shape & size
- Transparent, tissue-mimicking
- Replicating vessel behavior

Visualization Box

- Optically clear
- Sufficient space for flow development
- Enable visualization (e.g., PIV)

Connections

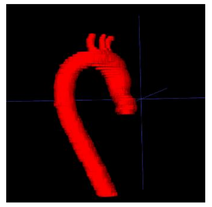
- Leak-proof interfaces
- Standardized fittings
- Easy assembly & modularity

Aorta Phantom

Aorta Geometry Reconstruction

Patient-specific anatomy was recovered from imaging in three stages: clinical CT slices provide the contrast data; segmentation in **ITK-SNAP** isolates the aortic lumen and wall; surface cleanup and meshing yield a closed **STL** solid suitable for SLA printing.

Input tomography; Hounsfield contrast defines soft tissue and lumen boundaries.



Label map from ITK-SNAP extracts the target vessel from the volume.

Post-processed surface mesh for slicing and stereolithography (SLA) printing.

Material Selection for Physiological Mimicry and Imaging

- **Formlabs 50A** resin was selected for PIV compatibility and flow visualization.
- It offers good resolution and optical access.
- Its elastic response enables partial simulation of pressure-induced radio expansion in cardiovascular tissue.

Figure 5 Phantom Fabricated with Formlab 50A Resin

Visualization Box

Purpose of Visualization Box

The visualization box was designed to apply external pressure to the aorta phantom and simulate physiological conditions. It also supports the phantom during testing.

Visualization Box Design

- **Operating pressure: 20 mmHg:**
  - Minimal wall deflection to avoid interference with PIV imaging .
- **Failure condition: up to 120 mmHg (systolic pressure)**
  - Structure must withstand pressure without failure.
- **Material selection: 3/8" polycarbonate**
  - Provides high optical clarity.
  - Ensures strong impact resistance.

Figure 6 Visualization Box

Fabrication

- **SLA printing** was used to fabricate the aorta phantom.
- Supports included during printing and removed after fabrication.
- Visualization box was assembled by **chemically welding polycarbonate panels**

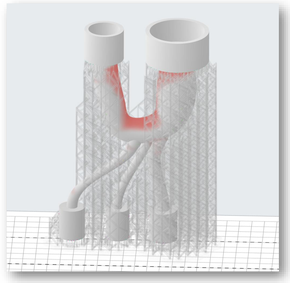


Figure 7 SLA Support Structures

Figure 8 SLA-Printed Aorta Phantom

Design Assembly

- System includes aortic phantom, visualization box, and connection components.
- Modular design allows easy assembly and replacement.

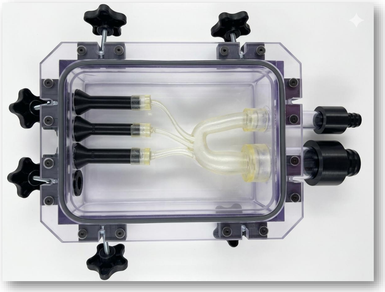


Figure 9 Physical Assembly Model

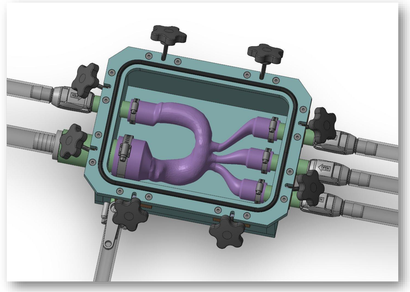


Figure 10 Digital Assembly Model

Future Work

1. Conduct **PIV flow visualization** experiments
2. Use the measured flow field to **validate PINN predictions**

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References

[1] "What is aortic stenosis?," Cleveland Clinic, https://my.clevelandclinic.org/health/diseases/23046-aortic-valve-stenosis (accessed Apr. 8, 2026).

[2] C. R. Smith *et al.*, "Transcatheter versus surgical aortic-valve replacement in high-risk patients," *New England Journal of Medicine*, vol. 364, no. 23, pp. 2187–2198, Jun. 2011. doi:10.1056/nejmoa1103510